



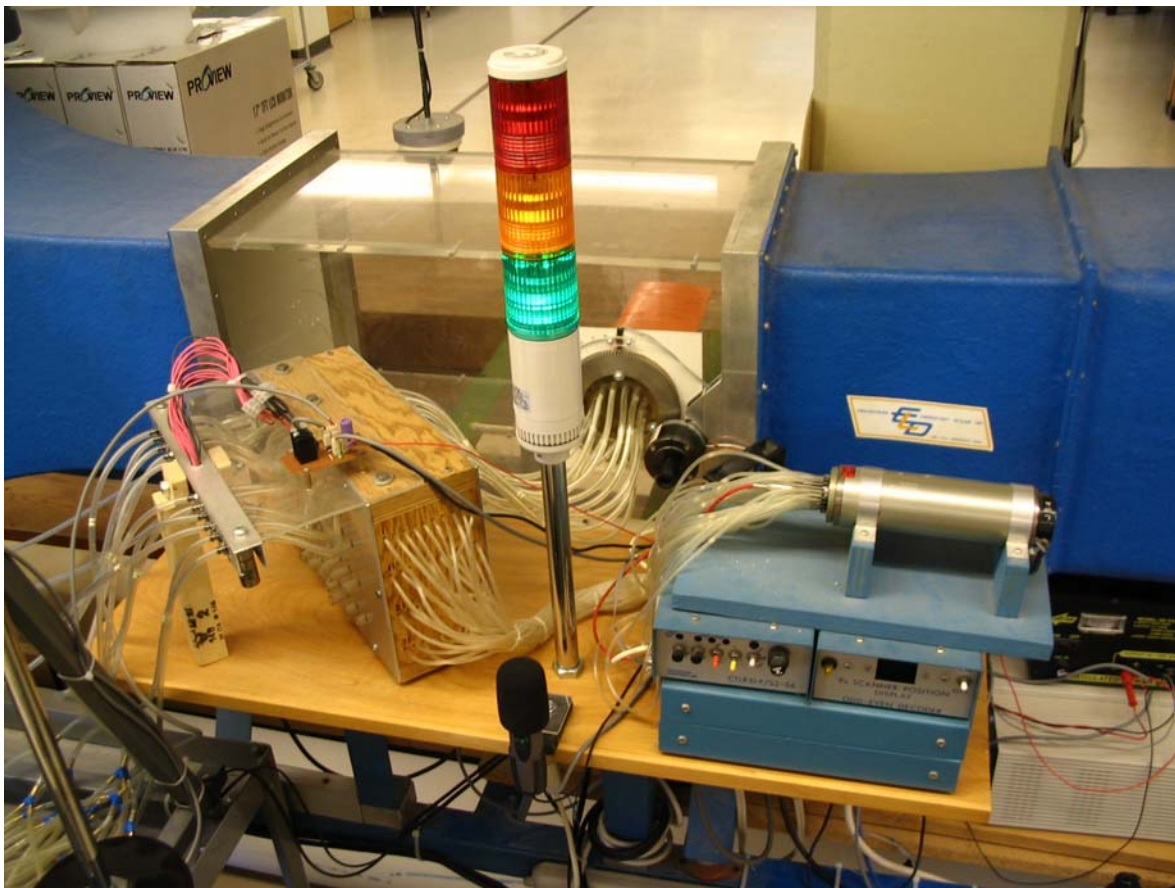
AER 303F Aerospace Laboratory I



Aerodynamic Forces on an Airfoil

<http://sps.aerospace.utoronto.ca/labs/raal>

Experiment Duration: 150 min



Instructor
M. R. Emami

Aerospace Undergraduate Laboratories
University of Toronto

Fall 2007

1. Purpose

Some basic concepts of subsonic flow are demonstrated using a nominal 50 m/s wind tunnel. The section-lift and drag coefficients for a symmetric airfoil are obtained by analyzing the measured pressure distributions at several pressure taps on the airfoil surface. Some observations are also made for the airfoil pitching motion and for small perturbations. The airfoil spans the tunnel test section. A multi-tube water manometer board is used to monitor the surface pressures and provide a visual display of the dynamic changes associated with varying angle of attack. A computer-aided data acquisition system is used to collect and record pressure data from the pressure probes.

2. Apparatus

- Open-loop subsonic tunnel ($V = 50$ m/s) with 1 ft² test section.
- NACA 0015 symmetric airfoil with 10 cm chord and upper surface pressure taps.
- Multi-channel electric Scanivalve[®] system with pressure transducer.
- Multi-tube water manometer board with solenoid valves.
- Betz manometer pressure gage.
- Stepper motor and Hall-effect sensor for pitching the airfoil.
- Data acquisition system (CIO-DAS16/Jr & CIO-DIO24 ISA Boards).
- Power supplies and power control unit.
- Web cams and audio system.

3. Notation and Constants

α	angle of attack (deg.)
α_{stall}	stall angle (deg.)
ds	a differential segment of the airfoil surface
c	chord length (cm)
C_d	drag coefficient
C_L	lift coefficient
C_p	pressure coefficient
C_{PU}	upper pressure coefficient
C_{PL}	lower pressure coefficient
C_x	x force coefficients
C_z	z force coefficients
D	section drag
L	section lift
P	normal airfoil skin pressure (mmH ₂ O)
P_∞	free stream static pressure (mmH ₂ O)
P_T	free stream total pressure (mmH ₂ O)
ρ	air density = 1.225 kg/m ³ (dry air)
U_∞	free stream velocity (m/s)

4. Experiment Setup

The major elements of the open-loop subsonic wind tunnel facility are indicated in Fig. 1. The tunnel operates as *open circuit*, meaning air is drawn from the laboratory and also exhausted to the laboratory.

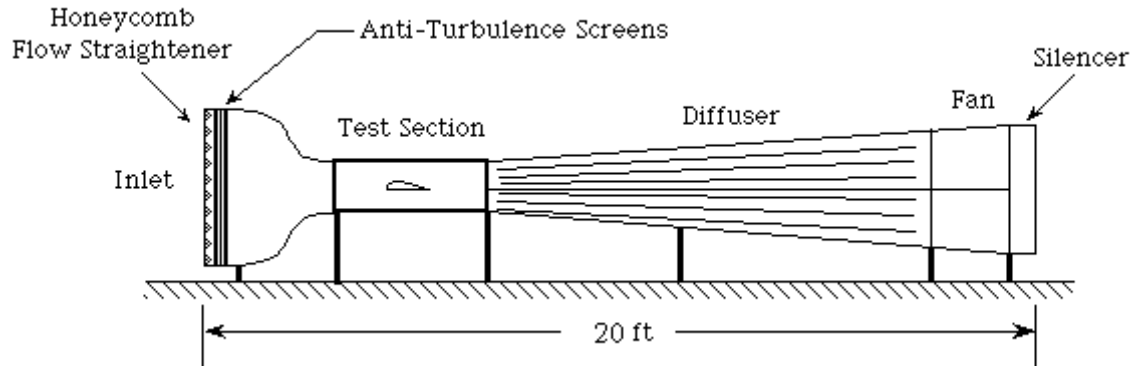


Figure 1: Schematic of the subsonic facility

4.1 Wind Tunnel

The power required to run a wind tunnel scales roughly as the third power of the flow velocity. This factor is reflected in the 15 HP (12 kW) motor required for starting and running at high speeds. Air is sucked through the test section by a large fan located at the rear of the tunnel. A cardboard honeycomb is employed in front of the test section to reduce flow turbulence and increase measurement accuracy.

A Pitot-static tube is mounted at the front of the test section to measure the static and impact pressures required to determine the flow velocity. The dynamic pressure from the Pitot-static tube is measured using a Betz manometer.

The laboratory wind tunnel incorporates a variable pitch NACA 0015 symmetric airfoil equipped with 11 surface pressure taps. A schematic of the airfoil pressure tap locations is depicted in Fig. 2. The tap coordinates are listed in Table 1.

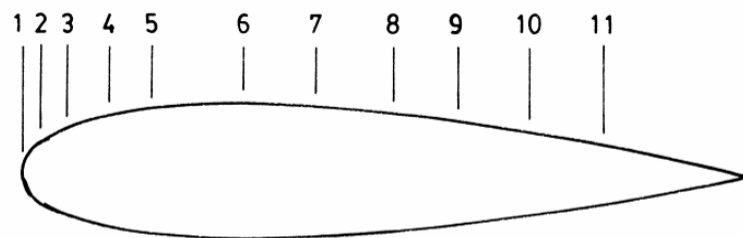


Figure 2: Pressure tap locations for the laboratory NACA 0015 airfoil

Table 1: Pressure tap locations

Tap	1	2	3	4	5	6	7	8	9	10	11
x (cm)	0.000	0.305	0.559	1.080	2.108	3.094	4.115	5.093	6.078	7.102	8.115

4.2 Data Acquisition System

The data acquisition system consists of a Pentium[®] II workstation equipped with CIO-DAS16/Jr and CIO-DIO24 data acquisition (DAQ) boards from Measurement Computing. The analog DAQ board (CIO-DAS16/Jr) has a 16-channel analog-to-digital (A/D) converter with 12-bit resolution (i.e., 2^{12} discrete voltage values over the measurement range of 0 to 10V), which is equivalent to a voltage resolution of 2.44 mV. The digital DAQ board (CIO-DIO24) has 24 digital I/O channels. The DAQ boards allow the computer software to control electromechanical actuators and collect data from a variety of analog and digital sensors.

The speed of the wind tunnel is set through a computer-controlled rotary potentiometer (driven by a DC motor) that converts the analog speed control knob position to a voltage that can be read by the DAQ board. Therefore, to change the speed of the wind tunnel the software moves the rotary potentiometer in small steps, while simultaneously reading the potentiometer voltage. Once the voltage corresponding to the desired speed is reached the software turns off the potentiometer motor.

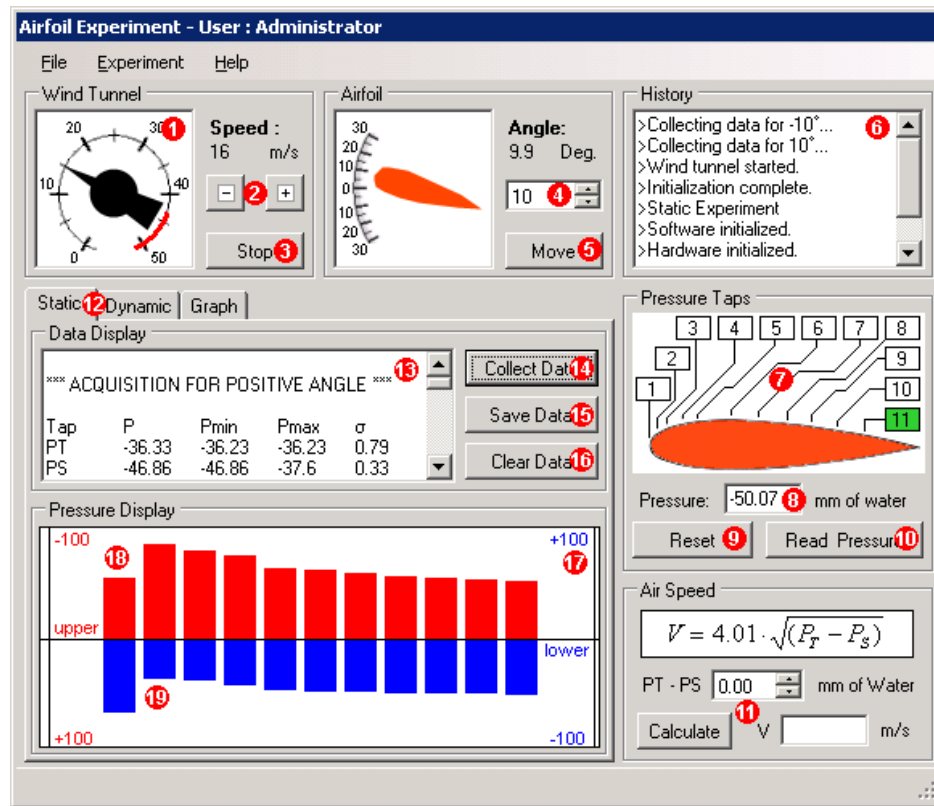
The pressure from the airfoil pressure taps is measured using a Scanivalve[®] pressure transducer system. The Scanivalve[®] sequentially connects each pressure tap to a pressure transducer that converts pressure values to a voltage that can be read by the DAQ board. Each pressure tap can also be connected to the water manometer board through an array of computer-controlled solenoid valves.

4.3 Airfoil Pitch Control

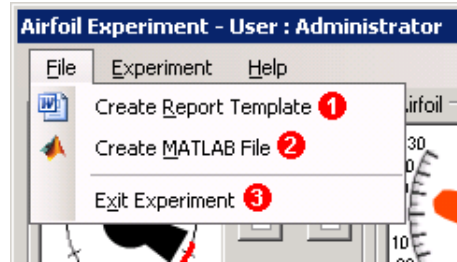
The angle of attack of the airfoil in the wind tunnel test section is changed using a computer-controlled pitching system that can rotate the airfoil through a full 360° . The pitching system consists of a stepper motor and gear transmission to rotate the airfoil with an angular resolution of 0.18° . A Hall-effect sensor is used for homing the stepper motor.

The computer software can control each component of the experiment individually, such as in the case of moving the airfoil to a desired angle of attack, or it can control multiple components simultaneously allowing complex dynamic experiments to be performed. The computer control system enables the experimenter to not only collect data more accurately, but also perform multiple tasks simultaneously.

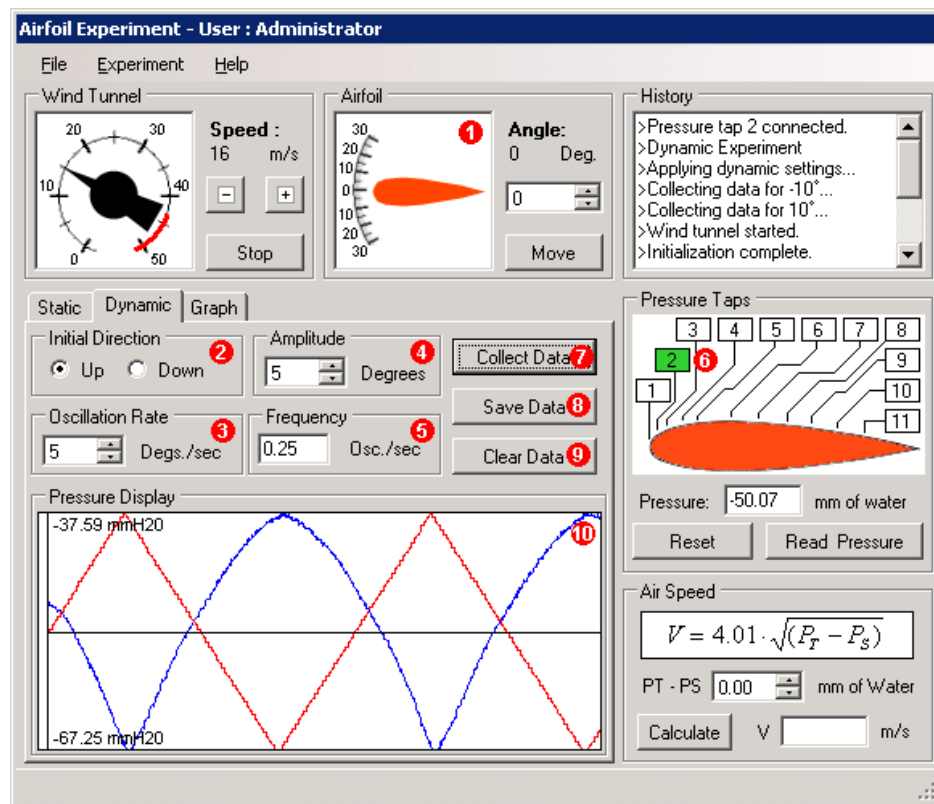
5. Experiment User Interface



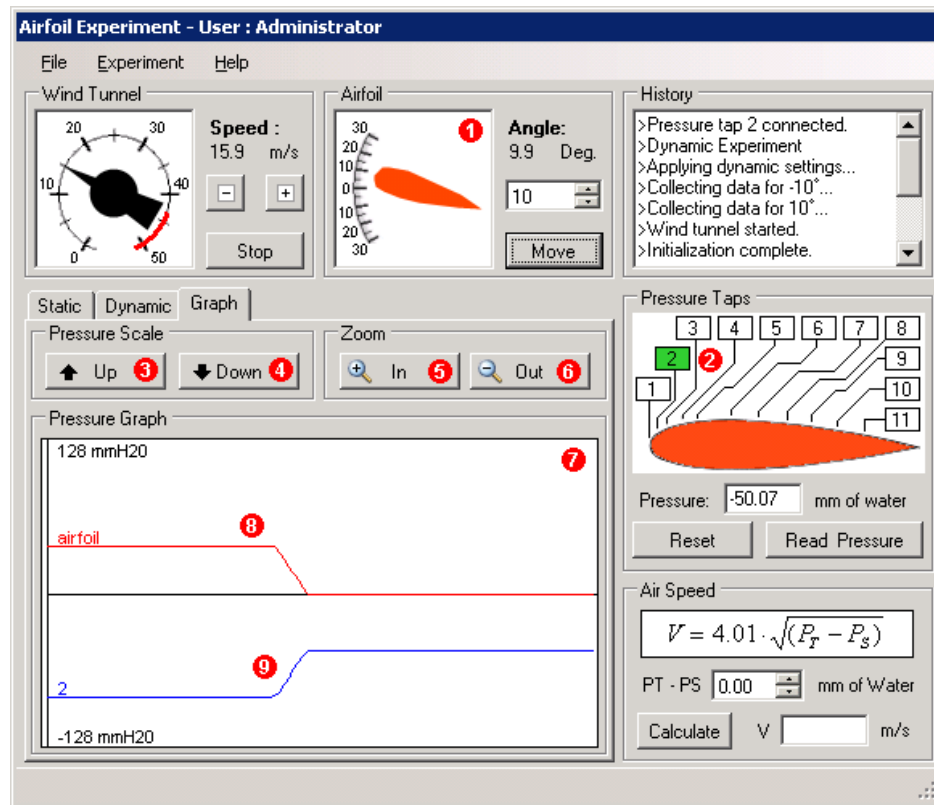
1. Graphic of the wind tunnel speed control knob.
2. Increase or decrease the speed of the wind tunnel. The displayed speed is only an *estimate*.
3. Start or stop the wind tunnel. There is a 60-second delay between starting and stopping the tunnel.
4. Select the angle of attack of the airfoil. The displayed angle is accurate to within 0.18°.
5. Move the airfoil to the selected angle of attack.
6. History of previously executed commands.
7. Graphic of the airfoil pressure tap locations. The currently connected tap is indicated in green. Click a pressure tap label to move the Scanivalve® to that tap.
8. Displays the measured pressure for the currently connected pressure tap.
9. Reset the Scanivalve® to the home position.
10. Measure the pressure for the currently connected pressure tap.
11. Calculate the air speed of the wind tunnel from the Betz manometer dynamic pressure. Enter the pressure from the Betz manometer and click *Calculate*.
12. Select between the *Static*, *Dynamic*, and *Graph* experiments.
13. Displays pressure data during a static experiment.
14. Record the airfoil pressure measurement for the selected angle of attack. Data is collected for both a positive and negative angle of attack automatically. All pressure measurements are in mmH₂O and are given *relative* to atmospheric pressure.
15. Save the static pressure data collected during an experiment.
16. Clear the data and pressure displays.
17. Displays pressure data during a static experiment as graphical bars.
18. Red bars indicate the pressure for the upper airfoil surface (positive angle of attack). Each bar corresponds to a pressure tap on the surface of the airfoil. The pressure scale (mmH₂O) is also in red on the left.
19. Blue bars indicate the pressure for the lower airfoil surface (negative angle of attack). Each bar corresponds to a pressure tap on the surface of the airfoil. The pressure scale (mmH₂O) is also in blue on the right.



1. Create a report template (.doc) for the experiment and save it to the Portal.
2. Create a MATLAB® file (.m) for the experiment and save it to the Portal.
3. Exit the experiment. Remember to first create a report template and MATLAB® file.



1. Select the angle of attack to oscillate about for the dynamic experiment.
2. Select the initial direction for the airfoil to pitch for the dynamic experiment.
3. Select the oscillation rate for the airfoil to pitch at for the dynamic experiment.
4. Select the peak amplitude of the pitch for the dynamic experiment. The peak-to-peak amplitude is twice the selected amplitude.
5. Displays the oscillation frequency of the airfoil based on the selected rate and amplitude.
6. Select the pressure tap on the airfoil to collect data at.
7. Record the airfoil angle of attack and pressure measurement for the currently connected pressure tap. Several oscillations will automatically be recorded. For low frequency oscillations with large amplitude this can take up to several minutes.
8. Save the dynamic pressure data collected during an experiment.
9. Clear the data and pressure displays.
10. Displays the airfoil angle of attack and pressure data *after* a dynamic experiment has been performed. The x-axis corresponds to the starting angle of the airfoil. The pressure scale (mmH₂O) is shown on the left.



1. Select the angle of attack for the airfoil.
2. Select the pressure tap on the airfoil to collect data at.
3. Move the pressure scale up.
4. Move the pressure scale down.
5. Zoom the pressure scale in.
6. Zoom the pressure scale out.
7. Displays live data of the airfoil angle of attack and the pressure for the currently connected pressure tap.
8. The angle of attack of the airfoil is displayed in red.
9. The pressure at the currently connected pressure tap is displayed in blue.

6. Subsonic Flow Theory

If the pressure distribution over the upper and lower surfaces of an airfoil is known then the section lift and drag coefficients (for inviscid flow) can be calculated.

The following statement gives a qualitative interpretation of the lift force forms on an airfoil. Consider a cross section of an infinite wing as depicted in Fig. 3, and a control volume specified by the surfaces **AB**, **BC**, **DE**, and **FG**. The air that passes through **AB** also flows through **DE**. Likewise, the air flowing through **BC** also travels through **FG**. From the diagram it can be seen that **DE** ≤ **FG**. By conservation of mass, the air flowing through **DE** must be moving faster than the air flowing through **FG**, providing **BC** = **AB**. Thus, the air must be flowing faster over the top of the airfoil than over the bottom surface. Therefore, according to Bernoulli's law the upper surface pressure must be less than that of the lower airfoil surface, resulting in an upward lift force.

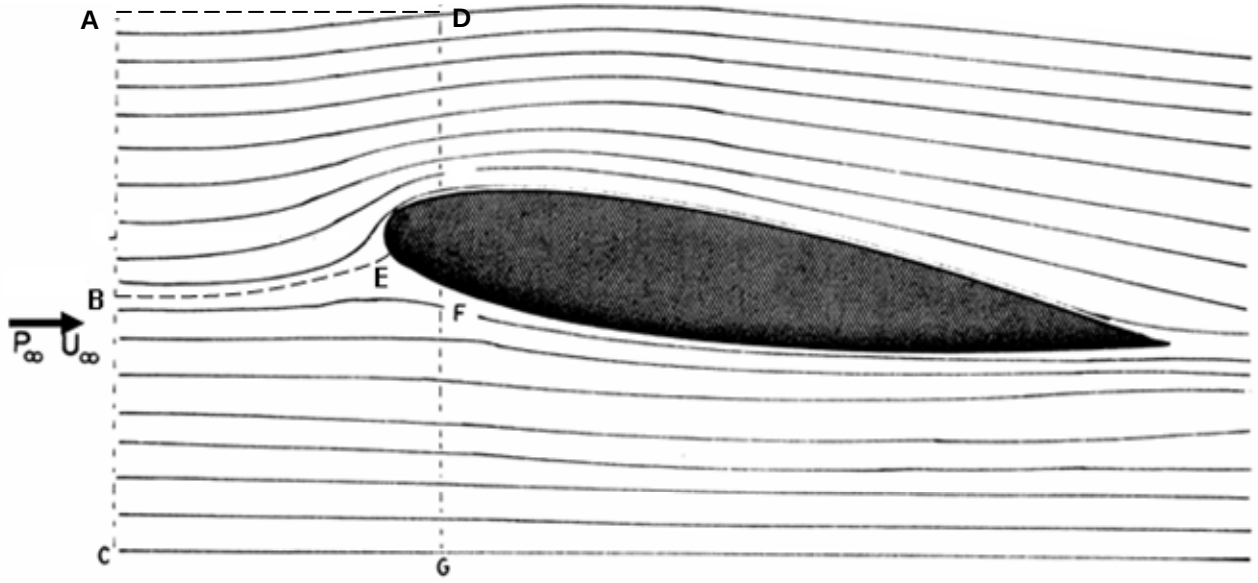


Figure 3: Streamlines around an airfoil in an incompressible flow

6.1 Pressure Coefficient

At any point in the flow where the local pressure is P the Pressure Coefficient C_P is defined as:

$$C_P = \frac{P - P_\infty}{\frac{1}{2} \rho U_\infty^2} \quad (1)$$

The Total or Stagnation Upstream Pressure P_T as measured by an impact probe (e.g., a Pitot tube) is the sum of the static and dynamic pressures at that point, i.e.,

$$P_T = P_\infty + \frac{1}{2} \rho U_\infty^2 \quad (2)$$

Thus, C_P may also be written in terms of differential pressures:

$$C_P = \frac{P - P_\infty}{P_T - P_\infty} \quad (3)$$

The presence of the airfoil in the test section will affect the test section velocity. For example, at a 15° angle of attack the local velocity over the airfoil will increase to about 1.02 times the upstream velocity due to blockage of the test section by the airfoil. This blocking effect will not be taken into consideration in the analysis that follows.

6.2 Lift and Pressure Drag

6.2.1 Normal Pressure Force

In Fig. 4 below, the normal pressure force Pds , acting on a small region, may be resolved into components dX and dZ acting parallel and perpendicular to the chord, respectively.

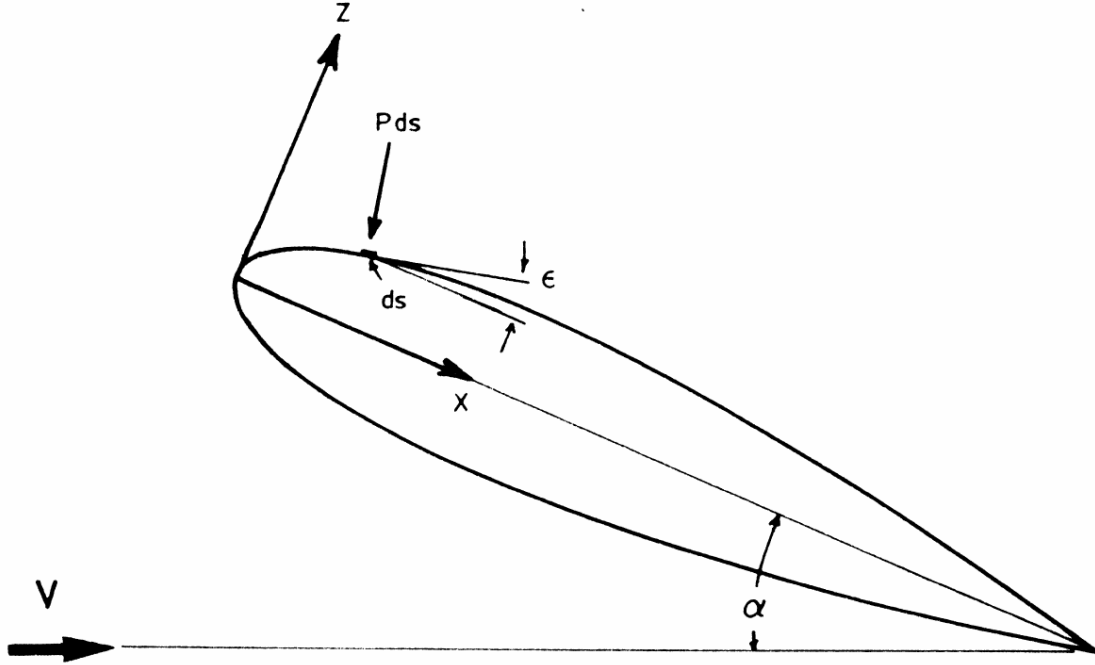


Figure 4: Normal pressure force on an airfoil surface

On the upper surface the z -component is given by:

$$dZ = -(P - P_\infty)_U ds \cos \epsilon \quad (4)$$

The normal force Z per unit span and chord is then given by:

$$dZ = -(P - P_\infty)_U dx \quad (5)$$

where $dx = ds \cos \epsilon$. Similarly, Z on the lower surface is give by:

$$dZ = (P - P_\infty)_L dx \quad (6)$$

Therefore, the total force in the z direction may be written as:

$$Z = -\int_0^c [(P - P_\infty)_U - (P - P_\infty)_L] dx \quad (7)$$

The normal force coefficient then becomes:

$$C_z = \frac{Z}{\frac{1}{2}\rho U_\infty^2 c} = -\int_0^1 (C_{PU} - C_{PL}) d\left(\frac{x}{c}\right) \quad (8)$$

6.2.2 Pressure Drag

The force coefficient parallel to the chord is responsible for pressure drag on the airfoil, and this coefficient may be derived in a manner similar to that previously employed for the normal pressure force.

First, divide the airfoil into 2 sections, fore and aft, such that fore is the area in front of the maximum thickness point of the airfoil as indicated in Fig. 5:

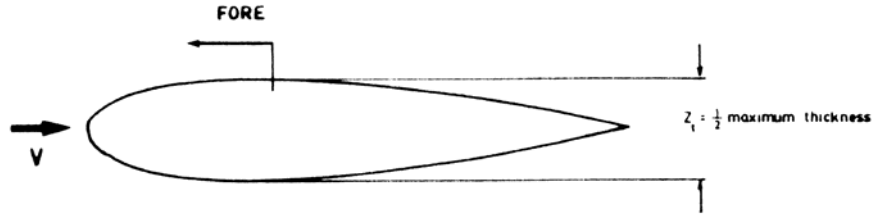


Figure 5: Pressure regions on an airfoil

For the fore section of the airfoil the x -component is given by:

$$dX = -(P - P_\infty)_U ds \sin \varepsilon \quad (9)$$

The force per unit span acting along the chord line is then given by:

$$dX = (P - P_\infty)_U dz \quad (10)$$

where $dz = ds \sin \varepsilon$. Similarly, for the aft section the x -component is given by:

$$dX = -(P - P_\infty)_A dz \quad (11)$$

Therefore, the total force per unit span in the x direction can be written as:

$$X = \int_{-z_t}^{z_t} (P - P_\infty)_F dz - \int_{-z_t}^{z_t} (P - P_\infty)_A dz \quad (12)$$

The corresponding force coefficient is then given by:

$$C_x = \frac{X}{\frac{1}{2}\rho U_\infty^2 c} = \int_{-\frac{z_t}{c}}^{\frac{z_t}{c}} C_p F d\left(\frac{z}{c}\right) - \int_{-\frac{z_t}{c}}^{\frac{z_t}{c}} C_p A d\left(\frac{z}{c}\right) \quad (13)$$

6.3 Lift and Drag Coefficients

Once the numerical values for C_z and C_x have been computed they may be used to calculate the lift and drag coefficients C_L and C_D using the following force relationships.

$$C_L = C_z \cos \alpha - C_x \sin \alpha \quad (14)$$

$$C_D = C_z \sin \alpha + C_x \cos \alpha \quad (15)$$

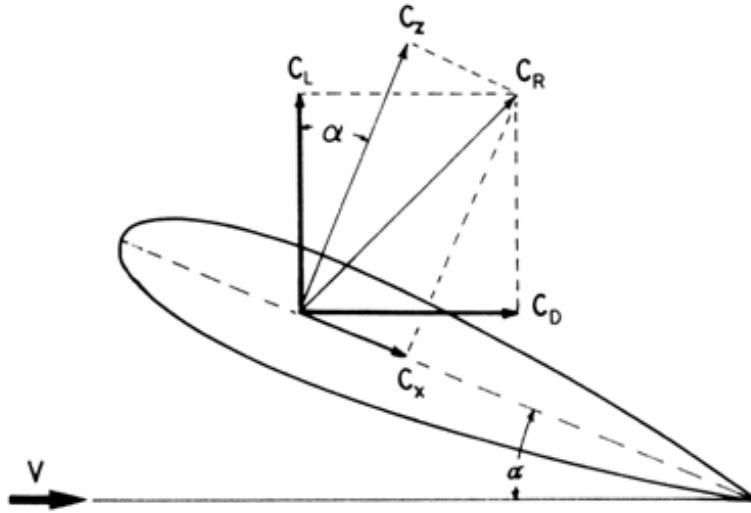


Figure 6: Component for diagram

6.4 NACA Airfoil Classification

For a four-digit NACA symmetrical airfoil the thickness to chord ratio is given by the following equation [1]:

$$\pm \frac{z}{c} = \frac{t}{0.2} \left[0.2969 \sqrt{\frac{x}{c}} - 0.126 \left(\frac{x}{c} \right) - 0.3516 \left(\frac{x}{c} \right)^2 + 0.2834 \left(\frac{x}{c} \right)^3 - 0.1015 \left(\frac{x}{c} \right)^4 \right] \quad (16)$$

where t is the maximum thickness of the section as a fraction of the chord dimension. In the case of the NACA 0015 airfoil the first two zeros imply a symmetrical section, and the number 15 represents the maximum thickness-to-chord ratio, for this case 15%.

7. Experiment Design

Some preparation and research will be required to understand your experiments prior to actually performing the tests in the wind tunnel. Each experiment should not be viewed as an independent activity. The results of one experiment *may* prove useful in defining the parameters of another test.

A MATLAB® file, named as “airfoil.m,” can be generated by the interface to plot the results of the Static Pressure Experiment. In order to run the function, use the following command in MATLAB® workspace:

```
>> airfoil ('.\<name of the folder where the data are saved>\<filename>.txt')
```

Note that the working directory must be the Airfoil folder. If you type in “help airfoil” in MATLAB® workspace you will find a brief description of the function. It is recommended, however, that you open the “airfoil.m” file and study the program and modify it for further analysis.

The tests to complete in this experiment are:

Static Pressure Measurement: To record the pressure at the taps along the surface of the airfoil use the controls under the “*Static*” tab. Set the air speed of the wind tunnel and move the airfoil to the desired angle of attack. Click the “*Collect Data*” button to collect pressure data for the selected angle of attack and for the inverse angle (negative angle). Click the “*Save Data*” button after an experiment to save the collected data in a text file.

Dynamic Pressure Measurement: To record the pressure at a single tap while the airfoil is pitching use the controls under the “*Dynamic*” tab. Set the air speed of the wind tunnel and move the airfoil to the desired starting angle of attack. Select the pressure tap to collect data from, the initial direction of the airfoil, the oscillation rate, and pitch amplitude. Click the “*Collect Data*” button to collect pressure and angle data for several periods of airfoil oscillation. Click the “*Save Data*” button after an experiment to save the collected data in a text file.

Pressure Variation Observation: To observe the change in pressure at a single tap in real-time use the controls under the “*Graph*” tab. Select the pressure tap to collect data from and adjust the scale and zoom so that the pressure display curve is visible. Moving the airfoil will cause the angle of attack and pressure curves to change in real-time.

7.1 Stall Angle Observations

Start the wind tunnel by clicking on “Start” button, and set the wind tunnel speed to approximately 25 m/s, using the “+” and “-” buttons. The air speed displayed on the interface is only an approximation. To obtain the exact speed of the flow use the dynamic pressure from the Betz manometer to calculate the air speed. A tool is provided in the experiment interface for performing this calculation.

Switch to the Pressure Graph by either clicking the “*Pressure*” tab or selecting “*Pressure Graph*” from the “*Experiment*” menu. Find the stall angle (α_{stall}) at each pressure tap by observing the pressure display at various angles of attack. You may need to zoom in/out to be able to see the pressure graph (blue curve). To change pressure taps click the number of the desired pressure tap on the “*Pressure Taps*” graphic. To change the angle of attack of the airfoil enter the desired angle in the “*Airfoil*” number box (or click on arrows) and click “*Move*”. In order to observe the pressure variation clearly you may need to adjust the graph scale by playing with up/down and zoom in/out buttons concurrently. Note that for this experiment the water manometer board is automatically disconnected from the pressure taps.

7.2 Static Pressure Experiment

(If you haven’t performed this already,) start the wind tunnel by clicking on “Start” button, and set the wind tunnel speed to approximately 25 m/s, using the “+” and “-” buttons. The air speed displayed on the interface is only an approximation. To obtain the exact speed of the flow use the dynamic pressure from the Betz manometer to calculate the air speed. A tool is provided in the experiment interface for performing this calculation.

Switch to the Static Pressure Experiment by either clicking the “*Static*” tab or selecting “*Static Pressure*” from the “*Experiment*” menu. Perform the **Static Pressure Measurement** for the following angles of attack: 0°, 3°, 6°, 9°, 12° and 15°. To change the angle of attack of the airfoil enter the desired angle in the “*Airfoil*” number box and click “*Move*”. Record $P_T - P_S$ from the Betz manometer. You can also observe the pressure distribution on the water manometer board. Do not forget to save your data for each angle of attack.

7.3 Dynamic Pressure Experiment

(If you haven’t performed this already,) start the wind tunnel by clicking on “Start” button (if you haven’t already started it), and set the wind tunnel speed to approximately 25 m/s, using the “+” and “-” buttons. The air speed displayed on the interface is only an approximation. To obtain the exact speed of the flow use the dynamic pressure from the Betz manometer to calculate the air speed. A tool is provided in the experiment interface for performing this calculation.

Switch to the Dynamic Pressure Experiment by either clicking the “*Dynamic*” tab or selecting “*Dynamic Pressure*” from the “*Experiment*” menu. For the Dynamic Pressure Experiment the water manometer board is automatically disconnected from the pressure taps to eliminate extraneous transients during high frequency pressure changes. Perform the **Dynamic Pressure Measurement** at taps 1 and 11 for each of the profiles listed in Table 2 for both “up” and “down” directions. The angle of attack (red) and pressure (blue) curves will appear in the display area after each test is completed. Do not forget to save your data after each test.

Table 2: Dynamic pressure experiment profiles

Profile	1	2	3	4	5	6	7	8	9	10
Amplitude	5°	5°	5°	10°	10°	10°	1°	1°	1°	1°
Rate (°/s)	1	5	10	1	5	10	1	5	10	20
Initial Angle	0°	0°	0°	0°	0°	0	0°	0°	0°, α_{stall}	0°, α_{stall}

8. Discussion of Results

8.1 Stall Angle Observations

1. What is your interpretation of the expression “The airfoil is stalled?”
2. How did you identify the stall angle from the real-time pressure graph?
3. How else would the stall condition manifest itself?
4. Tabulate the stall angle (α_{stall}) at each pressure tap.
5. For which pressure tap was stall most prominent? Why?

8.2 Static Pressure Experiment

1. From your raw pressure data, use the trapezoidal rule to calculate the lift and drag coefficients for the 9° case. Detail all the steps you employ in your computation.

Computation of C_x is a little more involved than that of C_z , because the intervals required for performing a good integration do not coincide with the tap intervals. The point of maximum airfoil thickness does not quite coincide with tap 6:

- At the correct integration limit $x/c = 0.300$ and $z/c = 0.07502$; but at tap 6 $x/c = 0.3094$.
 - A suitable interval for use with Simpson's or the trapezoidal rule can be obtained if the appropriate curves are plotted and then interpolated.
2. How are the recorded $P_T - P_S$ data from the Betz manometer compared with the Scanivalve[®] measurements (recorded in the data files)?

When checking the C_P values from the data files, remember that the calculations at tap 1 employ the average of the positive and negative angle of attack pressures.

3. For each angle of attack, plot the upper and lower pressure coefficients against chord-wise distance from the leading edge. Plot 3 sets of data per graph. Do not include error bars on these graphs.
4. On a single graph plot both C_L and C_D versus the angle of attack. Include error bars for both of the 9° data sets on this plot.
5. Plot C_D versus C_L ; this is known as the “polar” diagram.
6. Comment on all aspects of the experiment and be sure to indicate the Reynolds number on each of your plots.

7. By using the stream line sketches around airfoils, which are available in the literature, illustrate the flow patterns around the NACA 0015 airfoil at 0° , 9° and 12° angle of attack. Do you think the flow over the airfoil will be laminar or turbulent? Elaborate by considering various angles of attack. What method would you use for confirming laminar flow in the Static Experiment? Does the data obtained from your experiment agree with those reported in the literature? Explain.
8. Obtain polar graphs for other airfoils from the literature, and compare them with what you obtained in the Static Experiment.
9. Suggest some other methods that might be used to determine the lift of an airfoil. Try to use a few simple calculations to substantiate the reasons for your selections.

8.3 Dynamic Pressure Experiment

1. From the data for 10° amplitude, indicate and explain the conditions under which “turbulence” occurs in the flow. Explain the difference between the pressures measured at tap 1 and tap 11.
2. Do you observe significant difference between the data collected for the different initial pitch directions? Why?
3. Explain the differences between profiles with identical frequencies, but differing amplitudes.
4. From the data for small perturbations (Profile 7-10), is the flow over the airfoil laminar or turbulent? Explain.
5. Comment on any additional features of the data that you observed.
6. Discuss the major factors that cause turbulent flow in “airfoil pitching.” How would turbulence affect the forces on the airfoil?

9. References

- [1] Abbott, I. N., and Von Doenhoff, A. E., *Theory of Wing Sections*, Dover, 1959.
- [2] Anderson, J. D., *Introduction to Flight*, McGraw-Hill, 1978.
- [3] Goldstein (editor), *Modern Developments in Fluid Dynamics*, Dover, 1965.
- [4] Landon, R.H., *Oscillatory and Transient Pitching*, in AGARD Report: Compendium of Unsteady aerodynamic Measurements, AGARD-R-702, Data Set 3, 1982.
- [5] Barakos, G.N., Drikakis, D., *Unsteady Separated Flows Over Maneuvering Lifting Surfaces*, Philosophical Transactions: Mathematical, Physical, and Sciences, Vol. 358, No. 1777, pp. 3279-3291, 2000.